

Final examination

FULL NAME: _____

STUDENT ID: _____

Question	Max. points	Your points
1	7	
2	5	
3	10	
4	10	
5	10	
6	15	
7	8	
Total	65	

- Attempt all questions. Questions may appear on either side of a sheet.
- The exam is **180 minutes** in duration. The maximum score is **65 points**.
- The use of a non-programmable scientific calculator and an A4 sheet with handwritten information on both sides are allowed.
- Write your answers in this booklet. Request for extra paper only when needed.
- Sign the SUTD honour code in the back of this page. Good luck!



SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN HONOUR CODE

"As a member of the SUTD community, I pledge to always uphold honourable conduct. I will be accountable for my words and actions, and be respectful to those around me."

Introduction to the SUTD Honour Code

What is the SUTD Honour Code?

The SUTD Honour Code was established in conjunction with the school's values and beliefs, in good faith that students are able to discern right from wrong, and to uphold honourable conduct. It is an agreement of trust between the students and the staff and faculty of SUTD, and serves as a moral compass for students to align themselves to. Being in a university that aspires to nurture the leaders of tomorrow, it calls for students to behave honourably, not just solely in their academic endeavours, but also in everyday life.

What the Honour Code encompasses

Integrity & Accountability

To be honourable is to do what is right even when nobody is watching, and to be accountable for the things one does. One should always be accountable for one's words and actions, ensuring that they do not cause harm to others. Putting oneself in a favourable position at the expense of others is a compromise of integrity. We seek to create a community whereby we succeed together, and not at the expense of one another.

Respect

Part of being honourable is also respecting the beliefs and feelings of others, and to never demean or insult them. Should conflicts arise, the aim should always be to settle them in a manner that is non-confrontational, and try to reach a compromise. We will meet people of differing beliefs, backgrounds, opinions, and working styles. Understand that nobody is perfect, learn to accept others for who they are, and learn to appreciate diversity.

Community Responsibility

In addition to that, being honourable also involves showing care and concern for the community. Every individual has a duty to uphold honourable conduct, and to ensure that others in the community do likewise. The actions of others that display immoral or unethical conduct should not be condoned nor ignored. We should encourage each other to behave honourably, so as to build a community where we can trust one another to do what is right.

Student's signature

Q1) [7 points] Consider two machines, operating simultaneously and independently, where both machines have an exponentially distributed time to failure with mean $1/\mu$. There is a single repair facility, and the repair times are exponentially distributed with rate λ . What is the long run probability that no machine is operating?

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Q2) [2.5 + 2.5 = 5 points] Consider the continuous time Markov chain $(X_t : t \geq 0)$ on the state space of all integers $\mathbb{Z}$ with the following transition rates: $q_{i,i-1} = i^2 + 1$, $q_{i,i} = -2(i^2 + 1)$, $q_{i,i+1} = i^2 + 1$. Is the Markov chain recurrent? Is it positive recurrent? Justify your answers.

Q3)[6 + 4 = 10 points] A salesman flies around between Singapore, Bangkok, and Jakarta as follows. She stays in each city for an exponential amount of time with mean $1/2$ month if the city is Bangkok or Jakarta, but with mean $1/4$ month if the city is Singapore. From Singapore she goes to Bangkok or Jakarta with probability $1/2$ each; from Bangkok she goes to Singapore with probability $3/4$ and to Jakarta with probability $1/4$; from Jakarta she always goes back to Singapore.

- a) Find the limiting fraction of time she spends in each city.
- b) What is her average number of trips each year from Bangkok to Singapore in equilibrium?

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Q4)[10 points] In a communication system, messages requiring transmission arrive according to a Poisson process with rate λ . The messages are temporarily stored in a buffer having ample capacity. The buffer is empty at time $t = 0$. At the end of T time units, the buffer is cleared from all messages present. For each message there is a holding cost of $h > 0$ for each time unit the message has to wait in the buffer. Compute the expected total holding cost until time T in terms of the parameters λ, h, T .

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Q5)[10 points] Four teenagers enter a video game arcade which has two video game machines: the first one houses Speed Chess and the second one Street Fighter. One game of Speed Chess takes an average of 4 minutes to play, and is not very exciting; so when someone completes a game on it they always stand in line to play in the other machine (Street Fighter). One game of Street Fighter takes an average of 8 minutes to play, and is more interesting; so, upon completing one game, the person will get back in line to play it again with probability $1/2$ or go to play Speed Chess with probability $1/2$. Assuming that each complete game the teenagers play takes an exponentially distributed amount of time to finish (and are independent of each other), find the stationary distribution of the number of teenagers playing or in line at each of the two machines.

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Q6) [4 + 5 + 6 = 15 points] For a certain technology start-up, starting at a given time 0, patents are awarded as a Poisson process $N_p(\cdot)$ with rate 2 per quarter (three months), while lawsuits are filed against them as an independent Poisson process $N_l(\cdot)$ with rate 0.5 per quarter. *For parts (a) - (c) below, it is sufficient to provide expressions which can later be computed with a calculator.*

- a) Find the probability that, starting from time 0, in a period of one year, the combined total of patents awarded and lawsuits filed against is at most 3.
- b) Find the probability that, yet again starting at time 0, at least four patents are awarded before a total of two lawsuits are filed.
- c) Find the probability that the number of lawsuits never exceeds the number of patents filed.

Hints: Considering the first 5 events in the combined process might help for Part (b). For Part (c), viewing under the lens of Markov chains might help.

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Q7) [$6 + 2 = 8$ points] Data centers alternate between “working” and “down.” There are many reasons why data centers can be down, but for the purpose of this problem we mention only two reasons: (i) a backhoe accidentally dug up some cable, or (ii) a software bug crashed the machines. Suppose that a data center that is working today will be down tomorrow due to backhoe reasons with probability $1/6$, but will be down tomorrow due to a software bug with probability $1/4$. A data center that is down today due to backhoe reasons will be up tomorrow with probability 1. A data center that is down today due to a software bug will be up tomorrow with probability $3/4$.

- (a) What fraction of time is the data center working?
- (b) What is the expected number of days between backhoe failures?

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